Route Wise — Market Requirements Document (MRD)

DOC: MRD-v2.6-GOLD | **TEAM:** Cyber Knight
STATUS: Approved & Verified (99 Tests)
AUDIENCE: Evaluators, Mentors & Judges

The 30-Second Elevator Pitch: Most navigation apps blindly suggest the shortest route, funneling traffic into severe junction gridlock. **Route Wise (Team Cyber Knight)** is an intelligent traffic advisory platform, live glassmorphic dashboard, and installable PWA. It evaluates multiple driving alternatives, penalizes junction bottlenecks using Adjusted ETA with an anti-flapping switch threshold, predicts 15-minute trends via public **ARIMA (1,0,1)** ML, correlates crowd-sourced incidents with TomTom speed-drop telemetry via 2-vote consensus, provides municipal authority recurrent hotspot triage, and guarantees corridor retention during 1-tap Google Maps handoff.

1. Problem Framing & 5-Why Root Cause Analysis

Validated Problem Statement: Daily commuters and couriers face 15-to-30-minute delays at junction bottlenecks in Tier-1/2 Indian cities (e.g. Tiruchirappalli), causing missed SLAs, excess fuel burn, and stress. Standard GPS algorithms over-weight physical distance and lack transparent trade-off explanations. Therefore, there is an urgent need for an explainable routing system that penalizes junction gridlock and validates crowd alerts.

5-Why Diagnostic Root Cause Tree	Analysis & Evidence
1. Why delayed?	Trapped in bumper-to-bumper junction gridlock during peak hours.
2. Why enter?	Standard GPS displays the junction route as the shortest path in kilometers.
3. Why shortest?	Incumbent APIs use opaque black-box weights without customizable junction delay penalties.
4. Why no detour?	Commuters receive no side-by-side trade-off explanations (e.g. '+2.5 km but 16m faster').
5. Root Cause	**Lack of an explainable engine combining live delay, ML forecasting, and verified crowd alerts.**

Real-World Validation: Trichy Peak Rush Hour (CBS ■ Thillai Nagar across Cantonment)

Navigation System	Distance	Junction Delay	Trip Time	Driver & Courier Outcome
Standard Navigation App (Shortest distance default)	10.0 km (Shortest)	**+18 mins** (HPO bottleneck)	**42 mins**	Late delivery, customer SLA breach, high fuel wasted idling in gridlock.
Route Wise (Cyber Knight) (Adjusted ETA + ARIMA ML)	12.5 km (+2.5 km)	**+2 mins** (Smooth flowing)	**26 mins (16m faster!)**	On-time delivery, clear trade-off rationale, fuel-efficient constant speed corridor.

2. Market Opportunity & Sizing (Assumptions Explicitly Marked)

TAM: $14.2 Billion Global GIS, Navigation & Fleet Logistics API Market (CAGR 12.4%) **Assumption: Worldwide logistics transition to API-driven dispatch.*	**SAM: $2.8 Billion** Urban Last-Mile Delivery & Commuter Routing Software Platforms **Assumption: 20% of urban fleets adopt dynamic junction-avoidance software.*	**SOM: $350 Million** Tier-1 & Tier-2 Indian Logistics Startups, Fleets & Commuters **Assumption: Initial focus on South Indian freight hubs (Trichy, Coimbatore, Chennai).*

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3. Target Customer Personas

Persona	Role & Context	Pain Points & Unmet Needs	Route Wise Value Delivered
Karthik (26) **Commuter / Courier**	On-demand delivery courier & daily 2W commuter in Trichy.	Stuck at unannounced signal failures; misses strict 8 deliveries/hr SLA.	Trichy junction presets, 0ms autocomplete, 1-tap Google Maps handoff.
Rajesh (38) **Fleet Dispatcher**	Oversees 45 regional light commercial delivery vans across TN.	Fuel budget overruns by 14% due to idling; GPS apps flap routes erratically.	Adjusted ETA with anti-flapping threshold; 60s live auto-refresh.
Insp. Sundaram (48) **Traffic Authority**	Municipal traffic police & smart city urban planner.	Flood of citizen complaints without location data; no recurrence tracking.	Authority Hotspot Dashboard with 7-day spatial recurrence audit log.

4. Commuter User Journey: Current Gridlock vs Proposed Route Wise Flow

Journey Stage	Current Commuter Friction (Without Route Wise)	Proposed Route Wise Experience	Impact
1. Trip Planning	Enters destination; app selects single shortest path with zero trade-off info.	Compares up to 3 candidate driving routes with transparent Adjusted ETA.	**100% transparency**
2. Junction Triage	Enters bottleneck blindly; signal breakdown discovered only after arrival.	15-minute ARIMA ML trend and verified community incident markers visible.	**Proactive avoidance**
3. Navigation	Opening Google Maps recalculates and reverts back to congested main arterial.	Decision-anchor waypoints (≤ 3 waypoints) preserve chosen corridor.	**Corridor fidelity**

5. Competitive Landscape & Value Proposition

Feature / Capability	Google Maps API	Mapbox Navigation	Waze Platform	Route Wise (Cyber Knight)
Configurable Scoring Weights	No (Fixed defaults)	Partial (Profiles)	No (Fixed defaults)	Yes **— Adjusted ETA formula**
Transparent 'Why' Rationale	No (Opaque path)	No (No explanation)	No (No explanation)	Yes **— Comparative plain-English tags**
15-Min ML Traffic Forecast	No (Live snapshot only)	No (Historical only)	No (Live alerts only)	Yes **— Public ARIMA (1,0,1) model**
Anti-Flapping Route Threshold	No (Erratic flipping)	No (Flips on 1s delta)	Partial (Basic dampening)	Yes **— max(180s, 10% ETA)**
Corridor-Preserving GMaps Handoff	N/A	No (Raw coords only)	N/A	Yes **— Decision-anchor waypoints**
Incident Verification Consensus	Proprietary / Closed	No incident reporting	Unverified noise	Yes **— 2-Vote + speed drop (>30%)**
Offline Demo Fail-Safe	No (Fails on network drop)	No (Paid offline packs)	No (Network required)	Yes **— Built-in dictionary + PWA**

6. Business Model & Strategic Risk Mitigations

Tiered Monetization Model	Strategic Risk & Verified Mitigation Matrix
• **Developer Free ($0/mo):** 1,000 queries/mo, 60 req/min limit, full dashboard access, offline mock mode. • **Growth Fleet ($0.003/query):** Up to 250k queries/mo, custom scoring profiles, 60s live refresh API, ARIMA ML forecast. • **Municipal / Enterprise SLA:** Unlimited queries, Authority Hotspot Dashboard, 99.9% uptime SLA.	• **API Quota / Network Outage:** Local coordinate dictionary ensures 100% demo uptime. • **Spam Crowd Reports:** 2-vote peer consensus + TomTom speed-drop (>30%) correlation. • **Route Reversion on Handoff:** Decision-anchor waypoints (≤ 3 waypoints) preserve corridor. • **Database Contention:** Native Node 24 SQLite in Write-Ahead Logging (`WAL`) mode.

Route Wise — Product Requirements Document (PRD)

DOC: PRD-v2.6-GOLD | **TEAM:** Cyber Knight
STATUS: Approved & Verified (99 Tests)
AUDIENCE: Engineers, Mentors & Evaluators

Product Vision: Deliver an intelligent, traffic-aware routing platform and installable PWA that eliminates junction gridlock through transparent multi-criteria ranking, short-term ML forecasting, community consensus, and corridor-preserving navigation handoff.
Strategic Goals: G-1: Sub-450ms live API response with 100% demo availability. **G-2:** Anti-flapping route stability via timeSaved >= max(180s, 10% ETA). **G-3:** Verified incident overlay via 2-vote consensus + speed-drop correlation. **G-4:** Corridor preservation in Google Maps via anchor waypoints. **G-5:** Native mobile PWA support.
Non-Goals: NG-1: Proprietary audio turn-by-turn navigation engine. **NG-2:** Physical traffic light actuator hardware.

1. End-to-End User Experience & 7-Step Real-Time Interaction Flow

Step	Interaction Phase	System Action & Technical Detail	Trichy Real-World Example (CBS ■ Thillai Nagar)
Step 1	**Clean Input**	App opens centered on Trichy (zoom 13). User selects corridor preset or types custom names. 0ms autocomplete.	Clicks 'CBS ■ Chathiram'; origin and destination populate instantly.
Step 2	**Multi-Ingest**	Queries TomTom Routing API for driving alternatives. Ingests distance (m), duration (s), and live traffic delay (s).	Ingests Route #1 (Arterial: 10km, +18m delay) & Route #2 (Bypass: 12.5km, +2m delay).
Step 3	**Overlay**	Fetches active incidents from SQLite WAL store. Evaluates speed-drop telemetry correlation and runs ARIMA (1,0,1).	Detects signal failure report at HPO Junction correlated with 72% speed slowdown.
Step 4	**Adjusted ETA**	Ranks routes by Adjusted ETA without double-counting. Enforces switch threshold: timeSaved >= max(180s, 10% ETA).	Route #2 saves 16m (>180s and >10%); promoted to Recommended #1 pick!
Step 5	**Display & HUD**	Renders color-coded polylines on Leaflet map. Displays obsidian glass cards, trade-off rationale, and 15m forecast HUD.	Commuter reviews Route #2 card: 'Fastest path — Avoids 16 mins of junction delay'.
Step 6	**GMaps Handoff**	Extracts decision anchors (≤ 3 waypoints) preserving Route Wise's chosen corridor in external Google Maps navigation.	Launches Google Maps following Collector Bypass instead of reverting to HPO gridlock.
Step 7	**Authority Triage**	Municipal dispatchers review 7-day recurrent congestion hotspots and resolve cleared incidents with officer audit logs.	Traffic police clears signal failure; status updates across all connected clients.

2. Functional Requirements Traceability (Epics 1 to 3: FR-1 to FR-6)

Req ID	Epic / Module	Functional Specification & Acceptance Criteria	Priority	Status
FR-1	**Smart Geocoder**	Converts plain text names to GPS coordinates via TomTom with instant 0ms offline dictionary fallback for 12 verified Trichy junctions.	**P0 (Must)**	**Complete**
FR-2	**Multi-Route Ingest**	Requests candidate driving alternatives from TomTom API with real-time live traffic delay metrics and GeoJSON geometry.	**P0 (Must)**	**Complete**
FR-3	**Adjusted ETA Ranking**	Ranks routes by Adjusted ETA without double-counting delay. Applies switch threshold: timeSaved >= max(180s, 10% ETA) preventing route flapping.	**P0 (Must)**	**Complete**
FR-4	**ARIMA (1,0,1) Forecast**	Integrates public open-source zemlyansky/arima WebAssembly ML model to forecast velocity trends and delay spikes across +15m, +30m, +45m.	**P1 (Should)**	**Complete**
FR-5	**Native SQLite Store**	Native Node 24 SQLite storage (node:sqlite) with Write-Ahead Logging (WAL mode), automated migrations, and spatial proximity indexes.	**P1 (Should)**	**Complete**
FR-6	**Community Reporting**	Allows commuters to report 6 incident types (congestion, roadblock, roadwork, procession, signal_failure, waterlogging). Merges duplicates within 100m.	**P1 (Should)**	**Complete**

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2. Functional Requirements Traceability (Cont. Epics 4 to 7: FR-7 to FR-13)

Req ID	Epic / Module	Functional Specification & Acceptance Criteria	Priority	Status
FR-7	**2-Vote Consensus**	Auto-verifies reports when ≥ 2 positive community votes arrive or when TomTom speed drop (>30%) correlates with location.	**P1 (Should)**	**Complete**
FR-8	**Leaflet Incident HUD**	Verified incidents display solid halo + checkmark badge; Unverified display dashed border + question mark. 1-tap voting popup.	**P1 (Should)**	**Complete**
FR-9	**Authority Triage**	Protected console (X-Authority-Role: admin) ranking active incidents with 7-day spatial recurrence count and 1-click resolution audit log.	**P1 (Should)**	**Complete**
FR-10	**GMaps Anchor Handoff**	Extracts decision anchors (≤ 3 waypoints) preserving Route Wise's chosen corridor in Google Maps with driving and 2-wheeler modes.	**P1 (Should)**	**Complete**
FR-11	**60s Auto-Refresh**	Visual 60-second countdown timer that automatically refreshes delay metrics and re-evaluates ranking. In-memory raster tile cache.	**P2 (Could)**	**Complete**
FR-12	**Progressive Web App**	W3C manifest (display: standalone, #0b0f19 theme) and Service Worker (sw.js) with Network-First navigation and offline shell caching.	**P2 (Could)**	**Complete**
FR-13	**Trichy Presets & UX**	6 compact corridor presets (CBS ■ Chathiram, etc.), one-click junction chips, 120 FPS decoupled typing, and clear input controls.	**P2 (Could)**	**Complete**

3. Technical Architecture: Decoupled 3-Tier Pipeline

Frontend Tier (Vite + React 18): Obsidian dark glass interface, Leaflet maps, Service Worker (sw.js), and decoupled 120 FPS typing.
Backend Tier (Node.js + Express): SHA-256 API key authentication, sliding rate limiter, Adjusted ETA ranking engine, and TomTom raster tile cache.
Persistence & AI Tier: Native Node 24 SQLite (node:sqlite) in WAL mode + public **ARIMA (1,0,1)** ML forecasting engine.

4. REST API Endpoints Specification

Endpoint	Method & Auth	Payload / Query Params	Response Payload & Status	Latency Target
/api/routes/recommend	POST X-API-Key req	{ origin, destination, weights? }	200 OK: Ranked routes array with GeoJSON, Adjusted ETA, delay, and trade-off rationale.	< 350ms (Live) < 30ms (Fallback)
/api/routes/refresh	POST X-API-Key req	{ routeIds: [...] }	200 OK: Fresh delay metrics and recalculated scores without re-geocoding locations.	< 250ms (Live) < 20ms (Fallback)
/api/routes/locations/search	GET Public	?query=trichy	200 OK: Typeahead array with 0ms verified junction matching and isJunction tag.	< 15ms
/api/reports	POST User Key	{ category, lat, lon, description }	201 Created: Incident record ID, merged status, and verification state.	< 50ms
/api/reports/:id/votes	POST User Key	{ vote: 'yes' | 'no' }	200 OK: Updated vote tally; auto-transitions to verified at 2 votes.	< 35ms
/api/authority/hotspots	GET Admin Guard	X-Authority-Role: admin	200 OK: Ranked active incidents with 7-day spatial recurrence count.	< 40ms
/api/authority/reports/:id/resolve	POST Admin Guard	{ officerNotes?: string }	200 OK: Resolved incident object with resolvedAt audit timestamp.	< 40ms
/api/health	GET Public	None	200 OK: System health, TomTom state, SQLite health, server uptime.	< 15ms

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5. Persistence Schema & Non-Functional Requirements (NFRs)

SQLite WAL Database Schema	Non-Functional Requirements (NFRs) & Target Specifications
• **reports:** id PK, category, lat, lon, description, status, votes_yes, votes_no, reporter_key, created_at • **votes:** id PK, report_id FK, user_key, vote, created_at, UNIQUE(report_id, user_key) • **incident_resolutions:** id PK, report_id FK, resolved_by, notes, resolved_at • **Indexes:** Spatial composite index on reports(lat, lon) for sub-ms bounding box queries. • **Caching:** 1-hour in-memory cache for raster tiles (1ms response); 5-minute geocode typeahead cache.	• **Latency & Performance:** p95 API response < 450ms; 0ms local geocode fallback; 120 FPS typing. • **Security:** Cryptographic SHA-256 API key hashing; secrets redacted from logs; sliding rate-limiting. • **Availability:** 100% demo availability via resilient coordinate dictionary; Service Worker offline cache. • **Usability:** WCAG 2.1 AA compliant contrast; Cumulative Layout Shift (CLS) < 0.05. • **Compliance:** Zero personal telemetry tracking; pseudonymous client keys stored in localStorage.

6. Quality Assurance & Automated Test Verification (99/99 Tests Passing — 100% Pass Rate)

Test Suite Module	Test Count	Coverage Scope & Verified Behaviors	Pass Rate
Backend Test Suite (Node test runner)	**75 Tests**	SQLite WAL migrations, 2-vote consensus engine, speed-drop correlation (>30%), Adjusted ETA switch threshold, municipal authority role guard, raster tile caching, route controllers.	**100% (75/75)**
Frontend Test Suite (Node test runner)	**24 Tests**	Google Maps decision-anchor waypoint extraction, URL length compliance (<2048 chars), pseudonymous user key generation, route card formatters, severity badges.	**100% (24/24)**
Total Verification	**99 Tests**	Full monorepo end-to-end integration verified; production Vite build passes in 1.8s.	**100% (99/99)**

7. Release Criteria & 2-Minute Live Demo Script for Evaluators

Definition of Done (DoD): 100% test pass (99/99 tests), production Vite build cleanly compiled, PWA service worker registered, and offline coordinate fallback functional.
2-Minute Live Demo Script for Judges & Mentors:
• **Step 1 (Open Dashboard & PWA — 20s):** Launch application. Show the clean idle state centered on Trichy, the obsidian dark glassmorphism, and explain that Route Wise is an installable PWA with offline shell caching.
• **Step 2 (Click Preset Corridor — 30s):** Click the compact CBS ■ Chathiram preset. Origin and destination populate instantly (0ms local geocoding). The system queries TomTom live telemetry and fetches multiple alternatives.
• **Step 3 (Explain Adjusted ETA & Trade-Offs — 40s):** Point out how Route #2 is recommended because it saves 16 minutes of junction delay despite being 2.5 km longer. Explain the anti-flapping threshold tag and point to the 15-minute ARIMA ML traffic forecasting HUD.
• **Step 4 (Community Verification, Handoff & Tests — 30s):** Highlight community incident markers (verified solid pulse vs unverified dashed). Click *'Start Navigation in Google Maps'* to show corridor preservation via anchor waypoints. Conclude with our **99 automated tests passed at 100%**.